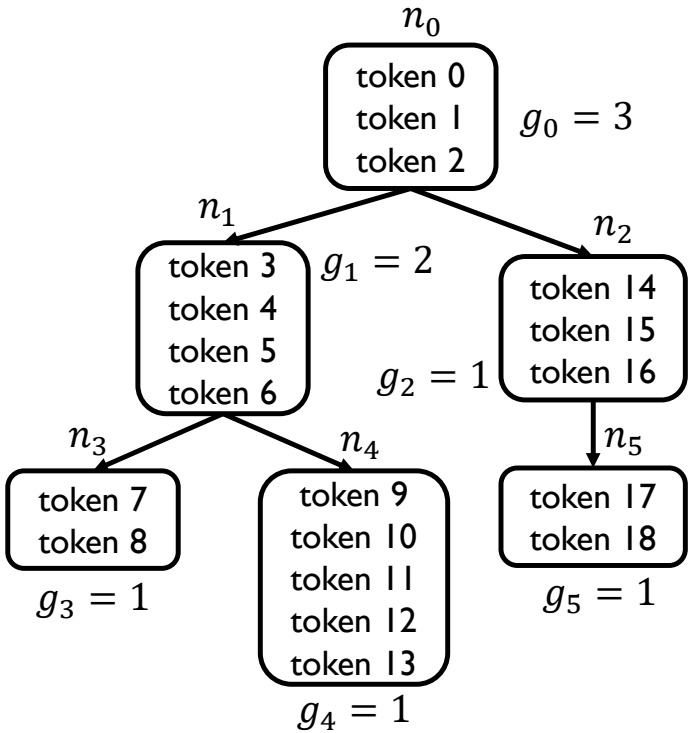


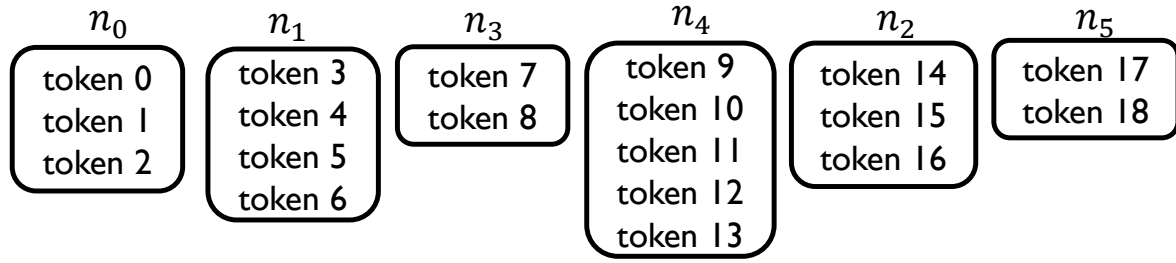
Trajectory tree (K=3, N=6)

Suppose there are K root-to-leaf path and N nodes in total.
A path k contains P_k nodes, a node n contains T_n tokens.

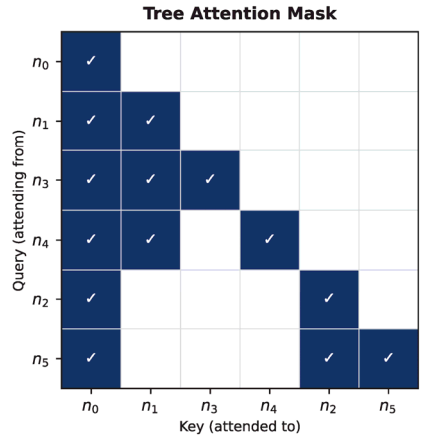
$$\mathcal{L}_{path_avg} = \frac{1}{K}\sum_{k=1}^{K}\sum_{n=1}^{P_k}\sum_{t=1}^{T_n}\ell_{k,n,t}(\theta)$$
$$= \frac{1}{K}\sum_{n=1}^{N} g_n \sum_{t=1}^{T_n}\ell_{n,t}(\theta)$$
$$= \frac{1}{K}\sum_{n=1}^{N}\sum_{t=1}^{T_n} g_{n,t}\,\ell_{n,t}(\theta)$$
$$= \sum_{n=1}^{N}\sum_{t=1}^{T_n}\frac{g_{n,t}}{K}\ell_{n,t}(\theta)$$

g_n is the number of path passing node n

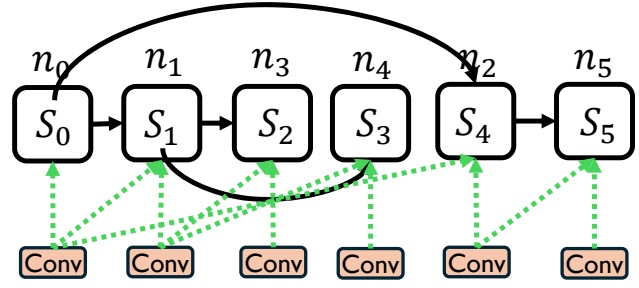
$g_{n,t}$ is the number of path passing a token in node n. $g_{n,t} = g_n$



DFS traversal



Tree attention mask



State and convolution routing